

Брунн–Минковский и смешанные объёмы

Homework

1 сентября 2026 года

1. Determine the equality cases in the box version of Brunn–Minkowski (Problem 1 of the class sheet), and deduce the equality case of the AM–GM inequality from it.
2. Prove that the additive form $|A + B|^{1/n} \geq |A|^{1/n} + |B|^{1/n}$ and the multiplicative form $|\lambda A + (1-\lambda)B| \geq |A|^\lambda |B|^{1-\lambda}$ are equivalent; explain which of them survives when $|A| = 0$.
3. Prove that equality in Brunn–Minkowski for convex bodies of positive volume forces A and B to be homothetic. (*Hint: settle $n = 1$ first, then induct on the dimension.*)
4. (Rogers–Shephard 1957) Prove $2^n|K| \leq |K - K| \leq \binom{2n}{n}|K|$, with equality on the left exactly for symmetric bodies and on the right exactly for simplices. (*Hint: on the left use Brunn–Minkowski; on the right compute $\int |K \cap (K + x)| dx$ in two ways.*)
5. (Urysohn) Prove $|K| \leq \omega_n(w(K)/2)^n$, where $w(K)$ is the mean width. (*Hint: average K over rotations.*)
6. Prove that mixed volumes are symmetric, multilinear, monotone and translation invariant, and that $nV(K, B, \dots, B)$ is the surface area of K .
7. State the Alexandrov–Fenchel inequality and deduce $W_i(K)^2 \geq W_{i-1}(K)W_{i+1}(K)$; verify the chain for a ball and for a segment.

8. (discrete analogue) Prove that for finite nonempty $A, B \subseteq \mathbb{Z}$ one has $|A + B| \geq |A| + |B| - 1$, with equality if and only if A and B are arithmetic progressions with the same common difference (or one of them is a single point). Compare with the case $n = 1$ of Brunn–Minkowski.
9. (VJIMC 1999, cat. II) Let A be a countable family of balls and Q a unit cube in $\mathbb{R}^3$, such that every finite subfamily can be packed into Q with disjoint interiors. Prove that the whole family can.
10. (Φ KH, becha 2024) A segment I *pierces* a sphere $S \subseteq \mathbb{R}^3$ if I meets S in two points (possibly coinciding). What is the maximal number of unit spheres with pairwise disjoint interiors that a segment of length 100 can pierce? (*Hint: the centres lie in a cylinder of radius 1 around the segment and are at pairwise distance at least 2; the key lemma is that the axial spread of any three such centres is at least $\sqrt{2}$.*)
11. (VJIMC 2000, cat. II) Let $\mathcal{B}$ be a family of open balls in $\mathbb{R}^n$ and $c < \lambda(\bigcup \mathcal{B})$. Prove that there are finitely many pairwise disjoint $U_1, \dots, U_k \in \mathcal{B}$ with $\sum_j \lambda(U_j) > c/3^n$.
12. Deduce the isoperimetric inequality for arbitrary compact sets (with Minkowski content in place of surface area) from Brunn–Minkowski, and point out where convexity was used in the class-sheet proof.